

Indian Statistical Institute, Bangalore

B. Math.

First Year, First Semester

Analysis I

Back Paper Examination

Total Marks 7x15=105

Maximum marks: 100

Instructor: B V Rajarama Bhat

Date : Dec. 22, 2025

Time: 3 hours

- (1) Let $\mathbb{Z}$ denote the set of integers. Show that $A := \{(m, n) : m, n \in \mathbb{Z}\}$ is countable. [15]
- (2) Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence of real numbers satisfying:

$$|a_{n+2} - a_{n+1}| \leq \frac{1}{3}|a_{n+1} - a_n|, \quad \forall n \geq 1.$$

- Prove or disprove: (i) $\lim_{n \rightarrow \infty} a_n$ exists. (ii) $\sum_{n=1}^{\infty} a_n$ converges. [15]
- (3) Let $\{b_n\}$ be a convergent sequence of real numbers. Show that

$$\limsup_{n \rightarrow \infty} b_n = \lim_{n \rightarrow \infty} b_n.$$

- (4) Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is a function. Define $f_1 : [0, 1] \rightarrow \mathbb{R}$ by [15]

$$f_1(x) = \begin{cases} f(x) & \text{if } f(x) \geq 0; \\ 0 & \text{otherwise.} \end{cases}$$

Show that: (i) If f is continuous then f_1 is continuous. (ii) Show that the converse of (i) is not true. [15]

- (5) Let $a, b \in \mathbb{R}$ with $a < b$. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function. Show that f is bounded. [15]
- (6) Let $g : [0, 1] \rightarrow \mathbb{R}$ be a function. Suppose g has local maximum at $\frac{1}{2}$ and g is differentiable at $\frac{1}{2}$. Show that $g'(\frac{1}{2}) = 0$. [15]
- (7) State and prove Rolle's theorem. [15]